

20	A	$I = E / \text{Total resistance}$

		$I = E / (R + r) \text{ ----- (1)}$ Since, E and R are constant, an <u>increase in r results in a decrease in current I.</u> Terminal pd $V = E - Ir \text{ ----- (2)}$ $V = E - [E / (R + r)] r$ Simplifying, we obtain $V = ER / (R + r)$ Since, E and R are constant, an <u>increase in r results in a decrease in pd V.</u>
21	B	Let current through $30\ \Omega$ resistor branch be I